



INDIAN INSTITUTE OF TECHNOLOGY ROPAR

Department of Mechanical Engineering

FINITE ELEMENT METHODS

(ME515)

Finite Element Analysis of a Cantilever Beam under Distributed Load

Submitted By

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Abstract

This project presents a detailed finite element analysis (FEM) of a cantilever beam subjected to a uniformly distributed transverse load using a two-dimensional plane stress formulation. The rectangular domain is discretized using four-node bilinear quadrilateral elements, and a computational code is developed to allow flexibility in geometry and mesh refinement. The study is divided into two major parts: static analysis and modal analysis.

In the static analysis, the displacement response, strain distribution, and stress fields are evaluated. A mesh convergence study is performed for a high aspect ratio beam to ensure solution accuracy, with displacement at a specified point and total strain energy used as convergence indicators. The numerical results obtained from FEM are validated against classical analytical solutions based on Euler–Bernoulli and Timoshenko beam theories. A parametric investigation is carried out for a wide range of aspect ratios to examine the influence of shear deformation and beam slenderness on structural response.

In the modal analysis, the natural frequencies and corresponding mode shapes of the cantilever beam are determined. Convergence of higher modes is also studied to ensure reliability of dynamic predictions.

The results demonstrate that the finite element model accurately captures both bending and shear effects, providing a smooth transition between classical beam theories. The developed framework serves as a robust tool for analyzing beam behavior under static and dynamic conditions.

Contents

1	Introduction	1
2	Objectives	1
3	Problem Description	2
3.1	Geometry and Boundary Conditions	2
3.2	Discretization	3
3.2.1	Mesh Parameters and Node Generation	3
4	Finite Element Formulation	5
4.1	Shape Functions	5
4.2	Coordinate Transformation	5
4.3	Displacement Approximation	5
4.4	Strain–Displacement Relation	6
4.5	Element Equations	6
4.6	Assembly of Global Stiffness Matrix	6
4.7	Global System of Equations	7
4.8	Analytical solutions	7
5	Mesh and Convergence Analysis	8
5.1	Mesh Refinement Strategy	8
5.2	Convergence Criteria	8
6	Results	9
6.1	Numerical Results	9
6.2	Error Analysis	9
6.3	Convergence Behavior	9
6.4	Error Convergence	10
6.5	Observations	10
7	Modal Analysis	11
7.1	Natural Frequencies	11
7.2	Frequency Convergence	11
7.3	Mode Shapes	12
7.4	Observations	12
8	Investigation of Beam Behavior for Varying the Aspect Ratios	13
8.1	Objective	13
8.2	Displacement vs Aspect Ratio	13

8.3	Strain Energy vs Aspect Ratio	13
8.4	Displacement Field	14
8.5	Strain Field	14
8.6	Stress Field	15
8.7	Shear Stress Across Thickness at $x = \frac{a}{10}$	15
9	Different Aspect Ratios Using ANSYS	16
9.1	Objective	16
9.2	Case 1: $a/b = 2$ ($a = 4$ m)	16
9.2.1	Observations (Aspect Ratio = 2)	16
9.3	Case 2: $a/b = 50$ ($a = 100$ m)	17
9.3.1	Observations (Aspect Ratio = 50)	17
10	Conclusion	18

1 Introduction

Cantilever beams are fundamental structural elements widely used in engineering applications and are often subjected to transverse loading. Accurate prediction of their deformation and stress distribution is essential for reliable design. Classical beam theories such as Euler–Bernoulli and

Timoshenko provide analytical solutions, but they have limitations in capturing detailed stress and shear effects, especially for beams with varying aspect ratios. The finite element method (FEM) offers a powerful numerical approach to overcome these limitations by discretizing the domain and solving the governing equations.

In this project, a two-dimensional FEM formulation using four-node quadrilateral elements is employed to analyze a cantilever beam under distributed loading. The study includes static analysis with convergence and validation, as well as modal analysis to determine natural frequencies and mode shapes.

2 Objectives

The main objectives of this project are as follows:

- Develop a finite element model of a cantilever beam using a 2D plane stress formulation with 4-node quadrilateral elements.
- Perform a static analysis of the beam under a uniformly distributed transverse load and evaluate the displacement, strain, and stress fields.
- Conduct a mesh convergence study and identify optimal discretizations for achieving accurate results.
- To validate the finite element results by comparing them with analytical solutions based on:
 - Euler–Bernoulli beam theory
 - Timoshenko beam theory
- Investigate the effect of aspect ratio on beam behavior through a parametric study.
- Perform modal analysis to determine natural frequencies and corresponding mode shapes of the cantilever beam.

3 Problem Description

A rectangular cantilever beam is considered in the xy -plane with length a and height b . The thickness t is small compared to the other dimensions, and plane stress conditions are assumed. The beam is subjected to a uniformly distributed transverse load q acting along the top edge.

3.1 Geometry and Boundary Conditions

A two-dimensional beam domain is considered with the following dimensions:

- Length $a = 100$ m
- Height $b = 2$ m
- Thickness $t = 0.01$ m

The beam has a high slenderness ratio:

$$\frac{a}{b} = 50 \quad (1)$$

The beam is made of Aluminum alloy, assumed to be homogeneous, isotropic, and linearly elastic.

- Young's modulus: $E = 200 \times 10^9$ Pa
- Poisson's ratio: $\nu = 0.3$
- Density: $\rho = 2700$ kg/m³

The beam is subjected to a cantilever boundary condition, where the left edge is fully fixed:

- $u_x = 0$
- $u_y = 0$

A distributed load q is applied along the top boundary, acting in the negative y -direction. The remaining boundaries are traction-free.

The vertical displacement is evaluated at point A, located at the midpoint of the bottom edge:

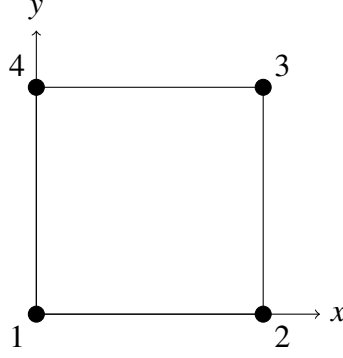
$$(x, y) = \left(\frac{a}{2}, 0\right) \quad (2)$$

The analysis is carried out under plane stress conditions, since the beam thickness is small and out-of-plane stresses are negligible.

$$D = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \quad (3)$$

3.2 Discretization

The rectangular domain is discretized into a finite number of four-node bilinear quadrilateral elements. The mesh consists of n_x elements along the x -direction and n_y elements along the y -direction, resulting in a structured grid.



3.2.1 Mesh Parameters and Node Generation

Let the domain be discretized into:

- n_x elements along the x -direction
- n_y elements along the y -direction

Total number of elements:

$$n_{el} = n_x \times n_y \quad (4)$$

Total number of nodes:

Since nodes are shared between adjacent elements,

$$N = (n_x + 1)(n_y + 1) \quad (5)$$

Element size:

The element dimensions are:

$$\Delta x = \frac{a}{n_x}, \quad \Delta y = \frac{b}{n_y} \quad (6)$$

Node coordinates:

Each node in the mesh is identified by indices (i, j) where:

$$i = 0, 1, 2, \dots, n_x \quad (\text{along } x\text{-direction}) \quad (7)$$

$$j = 0, 1, 2, \dots, n_y \quad (\text{along } y\text{-direction}) \quad (8)$$

The coordinates of a node are given by:

$$x_i = i \cdot \Delta x = i \cdot \frac{a}{n_x} \quad (9)$$

$$y_j = j \cdot \Delta y = j \cdot \frac{b}{n_y} \quad (10)$$

Global node numbering:

Nodes are numbered row-wise (left to right, bottom to top):

$$n(i, j) = j(n_x + 1) + i + 1 \quad (11)$$

Element connectivity:

Each quadrilateral element is defined by four nodes:

$$e(i, j) = \{n_1, n_2, n_3, n_4\} \quad (12)$$

where

$$n_1 = n(i, j) \quad (13)$$

$$n_2 = n(i + 1, j) \quad (14)$$

$$n_3 = n(i + 1, j + 1) \quad (15)$$

$$n_4 = n(i, j + 1) \quad (16)$$

Example:

For a mesh with $n_x = 4$ and $n_y = 2$:

$$n_{el} = 4 \times 2 = 8 \quad (17)$$

$$N = (4 + 1)(2 + 1) = 15 \quad (18)$$

This structured discretization ensures compatibility between elements and simplifies assembly of the global system.

4 Finite Element Formulation

4.1 Shape Functions

The shape functions for a four-node quadrilateral (Q4) element in natural coordinates (ξ, η) are:

$$N_1 = \frac{1}{4}(1 - \xi)(1 - \eta) \quad (19)$$

$$N_2 = \frac{1}{4}(1 + \xi)(1 - \eta) \quad (20)$$

$$N_3 = \frac{1}{4}(1 + \xi)(1 + \eta) \quad (21)$$

$$N_4 = \frac{1}{4}(1 - \xi)(1 + \eta) \quad (22)$$

4.2 Coordinate Transformation

The mapping between natural and physical coordinates is given by:

$$x = \sum_{i=1}^4 N_i x_i, \quad y = \sum_{i=1}^4 N_i y_i \quad (23)$$

The Jacobian matrix is:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \quad (24)$$

Derivatives are transformed as:

$$\begin{bmatrix} \frac{\partial N}{\partial x} \\ \frac{\partial N}{\partial y} \end{bmatrix} = J^{-1} \begin{bmatrix} \frac{\partial N}{\partial \xi} \\ \frac{\partial N}{\partial \eta} \end{bmatrix} \quad (25)$$

4.3 Displacement Approximation

The displacement field within an element is interpolated as:

$$\mathbf{u} = N\mathbf{d} \quad (26)$$

where

$$N = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix} \quad (27)$$

and $\mathbf{d}$ is the nodal displacement vector.

4.4 Strain–Displacement Relation

$$\boldsymbol{\varepsilon} = B\mathbf{d} \quad (28)$$

where the strain–displacement matrix B is:

$$B = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & \frac{\partial N_4}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial x} \end{bmatrix} \quad (29)$$

4.5 Element Equations

The element stiffness matrix is:

$$K_e = \int_{\Omega_e} B^T D B, d\Omega \quad (30)$$

The equivalent nodal force vector is:

$$F_e = \int_{\Gamma} N^T \mathbf{t}, d\Gamma \quad (31)$$

Numerical integration is carried out using 2×2 Gauss quadrature.

4.6 Assembly of Global Stiffness Matrix

The global stiffness matrix K is obtained by assembling the element stiffness matrices K_e over all elements in the domain.

Each element stiffness matrix is mapped to the global system using the connectivity between local and global node numbering. For a typical element with nodes (i, j, k, l) , the corresponding degrees of freedom are:

$$[u_i, v_i, u_j, v_j, u_k, v_k, u_l, v_l]$$

The assembly process is expressed as:

$$K = \sum_{e=1}^{n_{el}} A_e^T K_e A_e \quad (32)$$

where:

- n_{el} is the total number of elements
- K_e is the element stiffness matrix
- A_e is the connectivity (assembly) matrix relating local DOFs to global DOFs

The resulting global stiffness matrix K is a sparse matrix of size $2N \times 2N$, where N is the total number of nodes in the mesh.

After assembly, boundary conditions are applied by modifying the corresponding rows and columns of K and the force vector $\mathbf{F}$.

4.7 Global System of Equations

The global system is assembled as:

$$K\mathbf{d} = \mathbf{F} \quad (33)$$

Solving this system gives the nodal displacements.

4.8 Analytical solutions

$$A = t * b \quad (34)$$

$$I = \frac{t * b^3}{12} \quad (35)$$

$$G = \frac{E}{2(1 + \nu)} \quad (36)$$

$$\kappa = \frac{5}{6} \quad (37)$$

Displacement at $x = x_A$

$$v_{EB} = \frac{qx^2}{24EI} (6a^2 - 4ax + x^2) \quad (38)$$

$$v_{shear} = \frac{q}{\kappa GA} \left(ax - \frac{x^2}{2} \right) \quad (39)$$

$$v_{Timo} = v_{EB} + v_{shear} \quad (40)$$

Strain Energy

$$U_{EB} = \frac{q^2 a^5}{40EI} \quad (41)$$

$$U_{shear} = \frac{q^2 a^3}{6\kappa GA} \quad (42)$$

$$U_{Timo} = U_{EB} + U_{shear} \quad (43)$$

5 Mesh and Convergence Analysis

To ensure the accuracy and reliability of the finite element solution, a mesh convergence study is performed. The computational domain is discretized using structured quadrilateral elements, and the mesh is progressively refined until convergence is achieved.

5.1 Mesh Refinement Strategy

The number of elements in the x and y directions are denoted as n_x and n_y , respectively. The refinement process is carried out iteratively as follows:

- Start with an initial mesh (n_x, n_y)
- Increase mesh density in each iteration:

$$n_x^{(k+1)} = n_x^{(k)} + 50, \quad n_y^{(k+1)} = n_y^{(k)} + 1 \quad (44)$$

This strategy refines the mesh primarily along the beam length, which is critical due to the high aspect ratio.

5.2 Convergence Criteria

Convergence is monitored using two key quantities:

- Displacement at point A : $u_y(A)$
- Total strain energy: U

The relative error between successive iterations is computed as:

$$\text{Error in displacement} = \left| \frac{u_y^{(k)} - u_y^{(k-1)}}{u_y^{(k)}} \right| \quad (45)$$

$$\text{Error in energy} = \left| \frac{U^{(k)} - U^{(k-1)}}{U^{(k)}} \right| \quad (46)$$

The solution is considered converged when:

$$\text{Error} < 10^{-3} \quad (47)$$

6 Results

6.1 Numerical Results

The final converged results obtained from the finite element analysis are summarized below:

Quantity	Analytical	FEM
$u_y(A)$ (Euler–Bernoulli)	6.640625	6.612926
$u_y(A)$ (Timoshenko)	6.646475	
Strain Energy (EB)	7.500000×10^5	7.468073×10^5
Strain Energy (Timoshenko)	7.505200×10^5	

Table 1: Comparison of FEM and analytical results

6.2 Error Analysis

The percentage errors between FEM and analytical solutions are computed as:

Comparison	Error (%)
Displacement vs Euler–Bernoulli	0.4171
Displacement vs Timoshenko	0.5048
Energy vs Euler–Bernoulli	0.4257
Energy vs Timoshenko	0.4947

Table 2: Error comparison between FEM and analytical solutions

6.3 Convergence Behavior

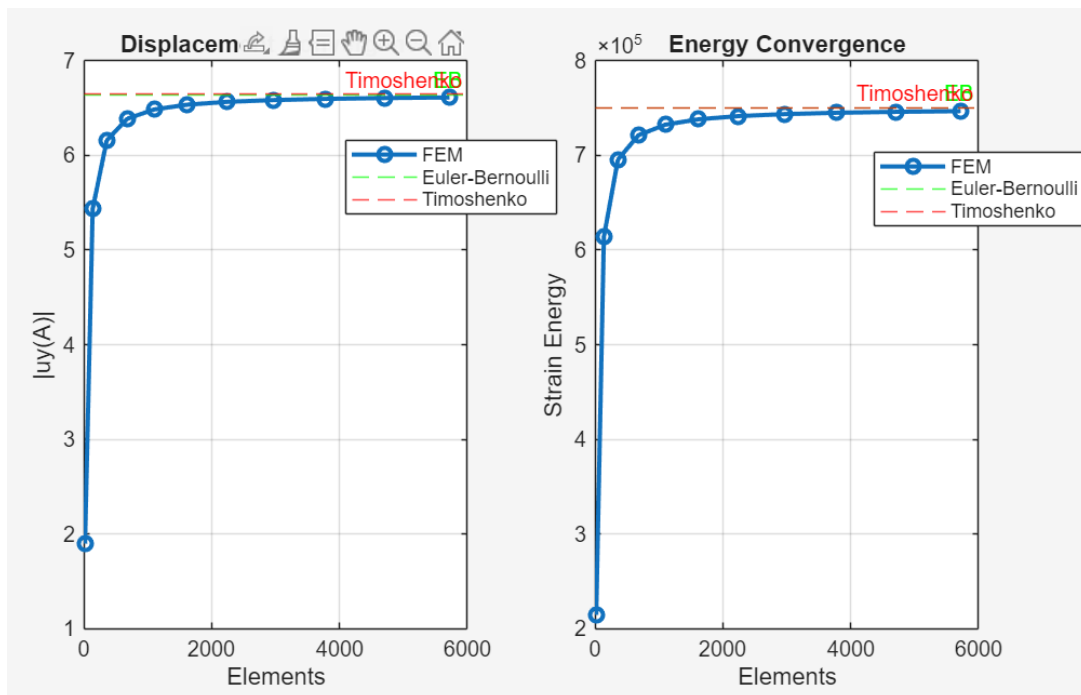


Figure 1: Convergence of displacement and strain energy with increasing number of elements

6.4 Error Convergence

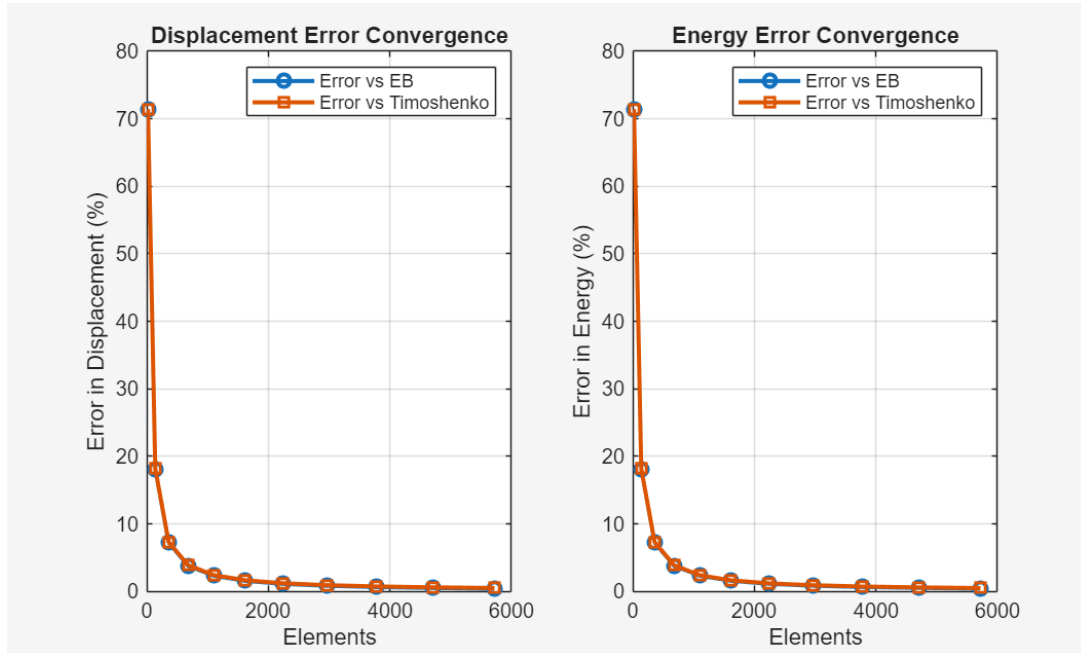


Figure 2: Convergence of percentage error in displacement and strain energy with mesh refinement

6.5 Observations

- The FEM solution converges rapidly with mesh refinement for both displacement and strain energy.
- The FEM results lie between Euler–Bernoulli and Timoshenko solutions, as it captures both bending and shear effects.
- The final error is very small (less than 1%), indicating high accuracy of the model.
- Error decreases sharply from coarse to fine mesh, showing strong convergence behavior.
- Displacement converges slightly faster than strain energy.
- The convergence is smooth and stable, confirming correct implementation of the FEM formulation.

Overall, the FEM model provides accurate and reliable results for the cantilever beam problem.

7 Modal Analysis

A modal analysis is performed to determine the natural frequencies and mode shapes of the cantilever beam. The governing eigenvalue problem is given by:

$$K\mathbf{d} = \omega^2 M\mathbf{d} \quad (48)$$

where K is the global stiffness matrix, M is the mass matrix, and ω is the Natural frequency.

The beam is fixed at the left end, and the eigenvalue problem is solved for the first five modes.

7.1 Natural Frequencies

The first five natural frequencies obtained from the finite element analysis are:

Mode Number	Frequency (Hz)
1	0.17244
2	1.0787
3	3.0121
4	5.8786
5	9.6674

Table 3: First five natural frequencies of the cantilever beam

7.2 Frequency Convergence

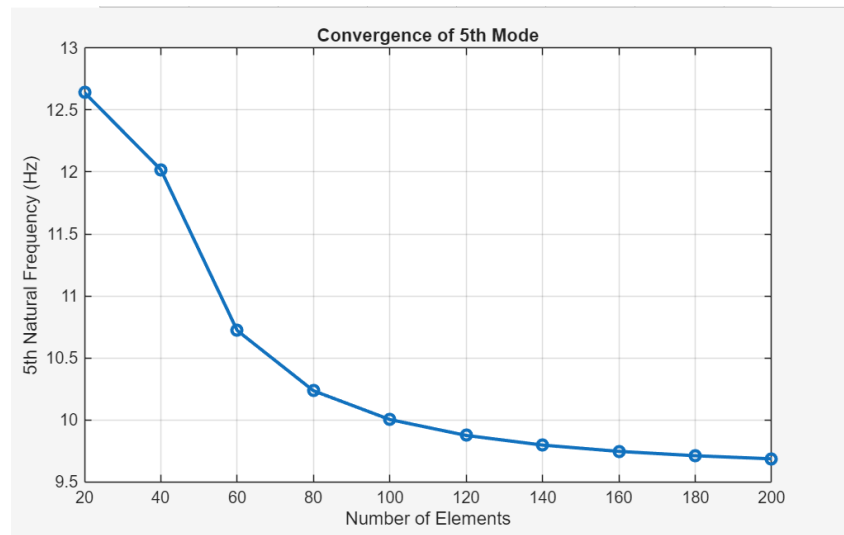


Figure 3: Convergence of the 5th natural frequency with mesh refinement

7.3 Mode Shapes

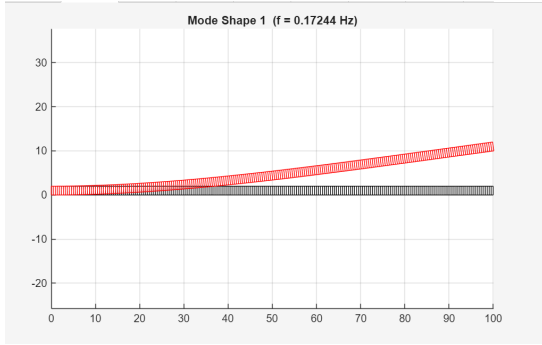


Figure 4: Mode 1: $f_1 = 0.17244$ Hz

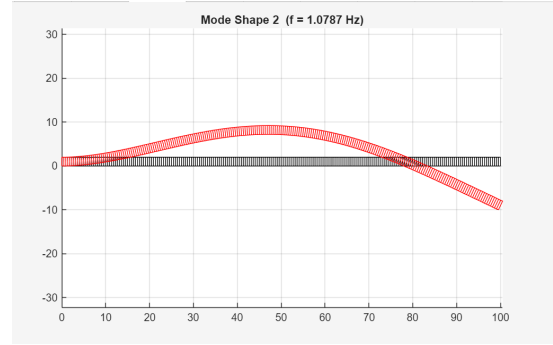


Figure 5: Mode 2: $f_2 = 1.0787$ Hz

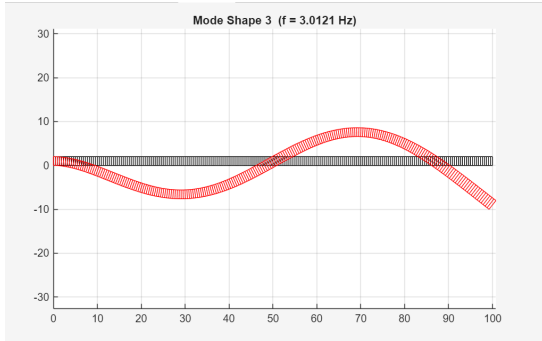


Figure 6: Mode 3: $f_3 = 3.0121$ Hz

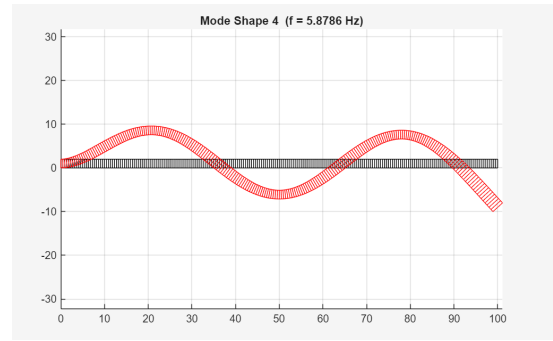


Figure 7: Mode 4: $f_4 = 5.8786$ Hz

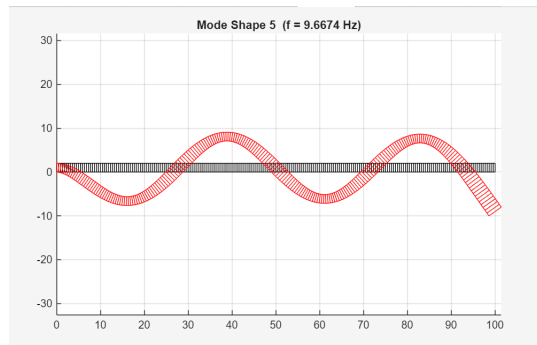


Figure 8: Mode 5: $f_5 = 9.6674$ Hz

7.4 Observations

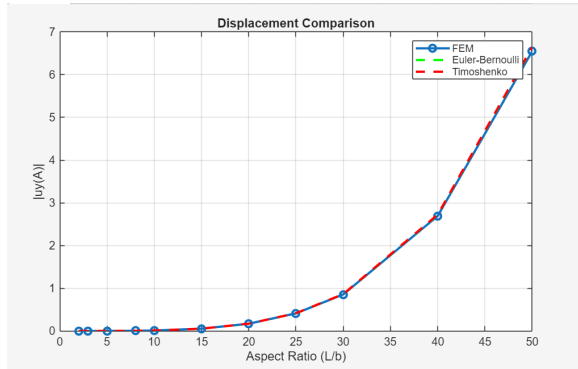
- The natural frequencies decrease and converge with mesh refinement, indicating stable FEM formulation.
- The 5th mode converges to approximately 9.75 Hz for finer meshes.
- Mode shapes show increasing number of curvature changes (nodes) with higher modes.

8 Investigation of Beam Behavior for Varying the Aspect Ratios

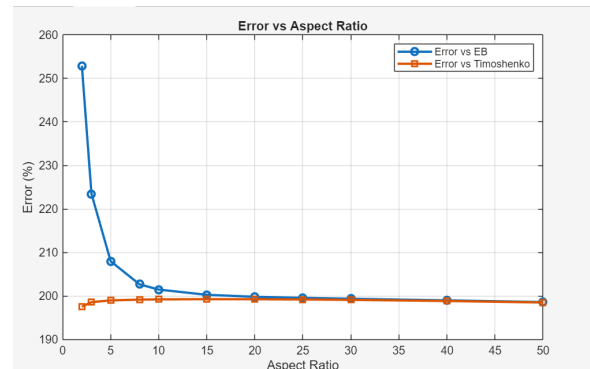
8.1 Objective

The objective of this investigation is to analyze the behavior of a beam using a converged finite element mesh for varying aspect ratios ranging from 2 to 50 (2, 3, 5, 8, 10, 15, 20, 25, 30, 40, 50). The FEM results are compared with analytical solutions obtained from Euler–Bernoulli and Timoshenko beam theories. Additionally, shear stress variation across the thickness at $x = \frac{a}{10}$ is studied, along with displacement, strain, and stress fields.

8.2 Displacement vs Aspect Ratio



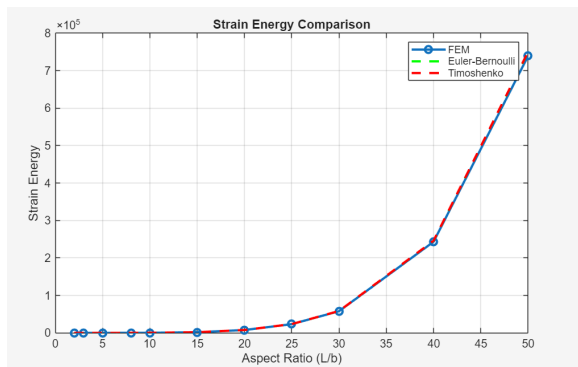
(a) Displacement vs Aspect Ratio



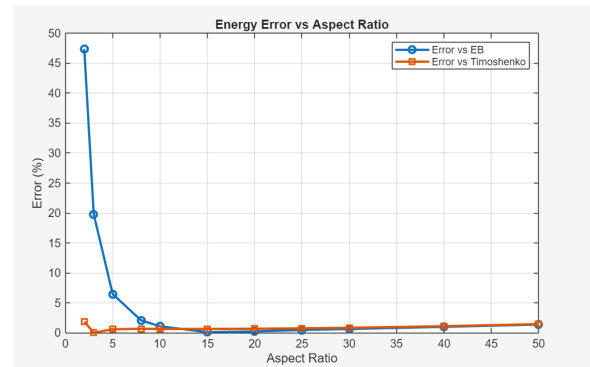
(b) Error vs Aspect Ratio

Figure 9: Displacement behavior of beam for varying aspect ratio

8.3 Strain Energy vs Aspect Ratio



(a) Strain Energy vs Aspect Ratio



(b) Energy Error vs Aspect Ratio

Figure 10: Strain energy variation with aspect ratio

8.4 Displacement Field

The displacement increases from the fixed end to the free end, with maximum deflection at the tip. The smooth variation confirms bending-dominated behavior and correct FEM implementation.

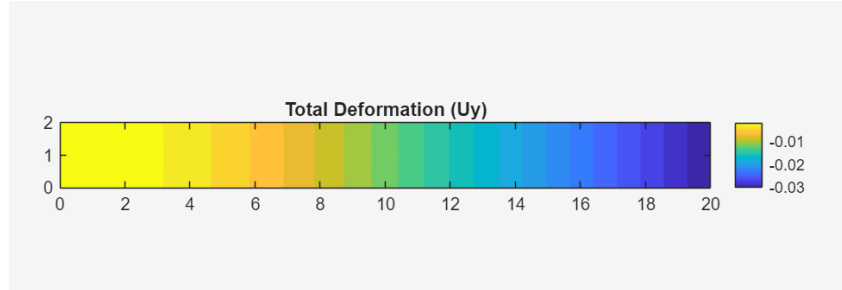


Figure 11: Displacement Field (u_y)

8.5 Strain Field

The normal strain ϵ_x varies linearly across the beam depth, with tensile strain at the top surface and compressive strain at the bottom surface.

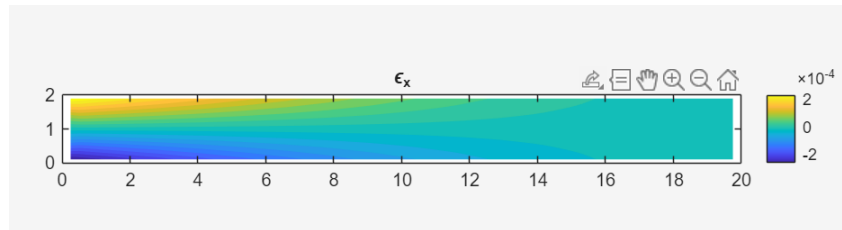


Figure 12: Normal Strain Field (ϵ_x)

The shear strain γ_{xy} is maximum near the mid-height of the beam and decreases towards the top and bottom surfaces.

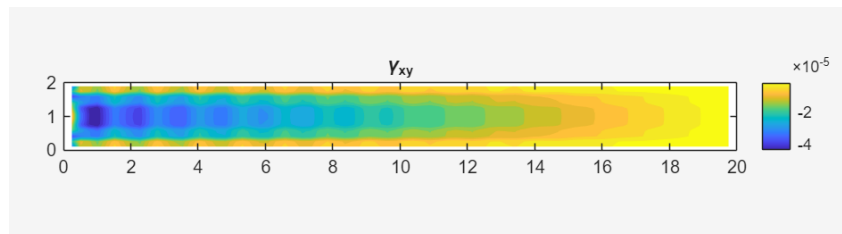


Figure 13: Shear Strain Field (γ_{xy})

8.6 Stress Field

The normal stress σ_x varies linearly across the beam depth, with tensile stress at one surface and compressive stress at the opposite surface.

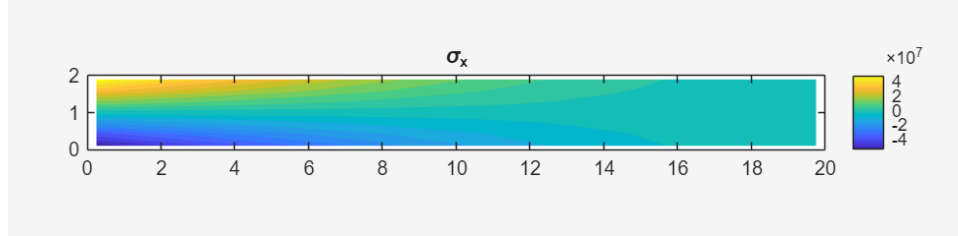


Figure 14: Normal Stress Field (σ_x)

The shear stress τ_{xy} is maximum near the neutral axis and decreases towards the top and bottom surfaces.

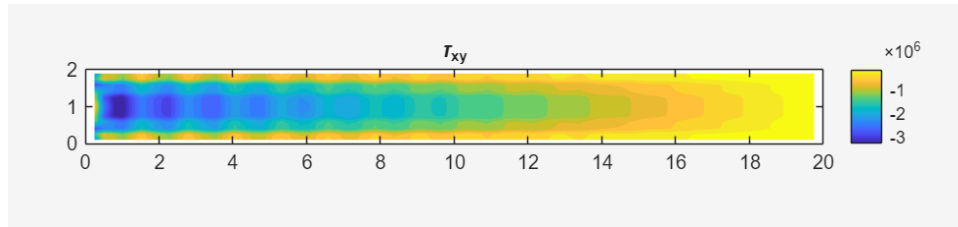


Figure 15: Shear Stress Field (τ_{xy})

8.7 Shear Stress Across Thickness at $x = \frac{a}{10}$

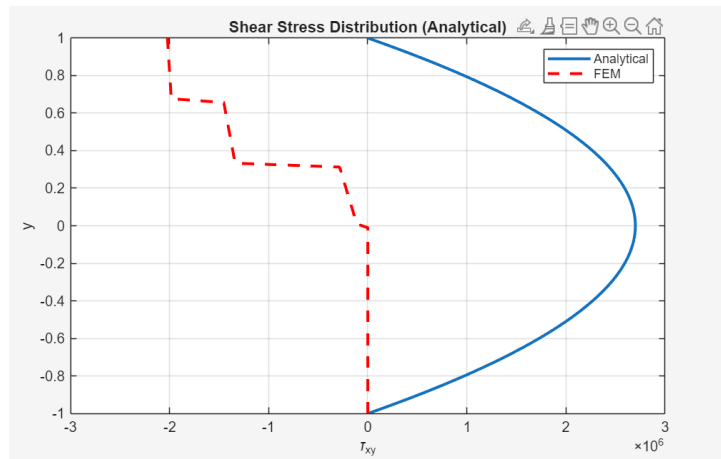


Figure 16: Shear Stress Across Thickness

The analytical shear stress shows a parabolic distribution, while the FEM result appears piecewise due to discretization. Both follow the same trend, with maximum at the neutral axis.

9 Different Aspect Ratios Using ANSYS

9.1 Objective

The objective of this study is to validate the FEM results using ANSYS for different aspect ratios ($L/H = 2, 50$) and analyze deformation, stress, and strain distributions.

- Height: $b = 2$ m, Thickness: $t = 0.01$ m
- Material: Structural Steel ($\rho = 7850 \text{ kg/m}^3$, $\nu = 0.3$)

9.2 Case 1: $a/b = 2$ ($a = 4$ m)

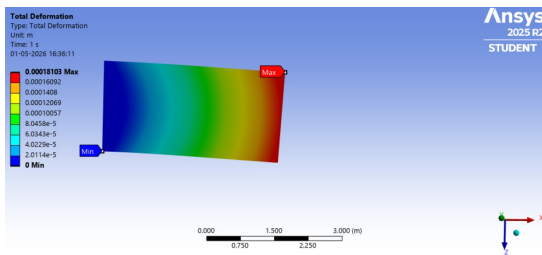


Figure 17: Deformation

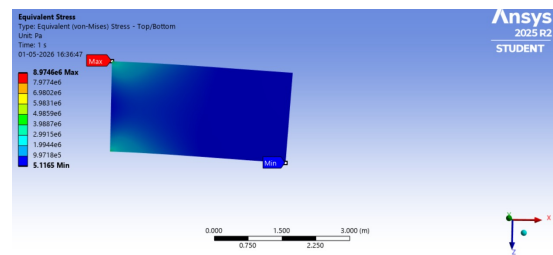


Figure 18: Stress

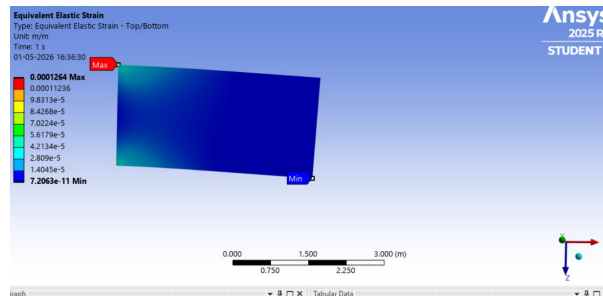


Figure 19: Strain

Figure 20: ANSYS results for $a/b = 2$

9.2.1 Observations (Aspect Ratio = 2)

- The displacement is relatively small, indicating high stiffness of the thick beam.
- The deformation is not purely bending; noticeable shear deformation is present.
- The equivalent strain is concentrated near the fixed end and spreads across the depth, showing combined bending and shear effects.
- The stress distribution is more uniform compared to slender beams, with significant shear contribution.

9.3 Case 2: $a/b = 50$ ($a = 100$ m)

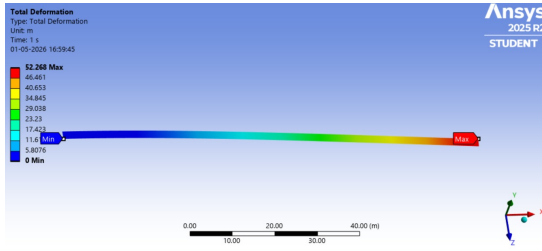


Figure 21: Deformation

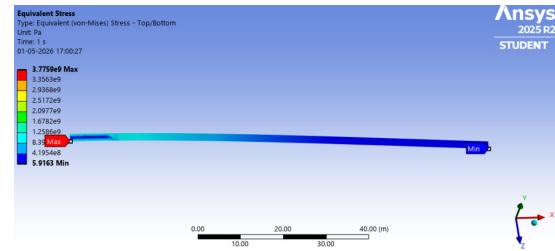


Figure 22: Stress

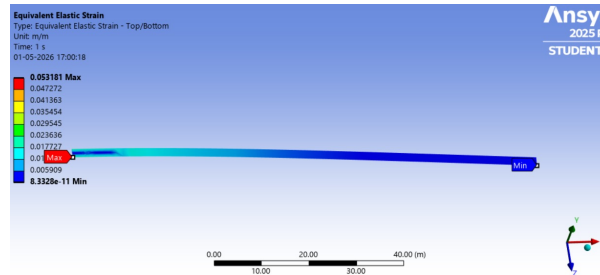


Figure 23: Strain

Figure 24: ANSYS results for $a/b = 50$

9.3.1 Observations (Aspect Ratio = 50)

- The displacement is significantly larger, indicating reduced stiffness of the slender beam.
- The deformation is predominantly bending, with negligible shear effects.
- The strain is highly concentrated near the fixed end and varies linearly across the depth.
- The stress distribution follows classical bending theory, with clear tension and compression zones.
- The maximum stress occurs at the fixed end and decreases smoothly along the beam length.

10 Conclusion

A comprehensive finite element analysis of a cantilever beam under distributed loading was carried out using a 2D plane stress formulation. The FEM results showed excellent agreement with analytical solutions from Euler–Bernoulli and Timoshenko beam theories, with errors less than 1%, confirming the accuracy of the implementation.

The mesh convergence study demonstrated stable and reliable numerical behavior, while modal analysis validated the dynamic response through converged natural frequencies and mode shapes.

The parametric study on aspect ratio revealed that beam behavior transitions from shear-dominated (thick beams) to bending-dominated (slender beams). For low aspect ratios, shear effects are significant and Timoshenko theory provides better predictions, whereas for high aspect ratios, Euler–Bernoulli theory becomes sufficient.

The stress, strain, and displacement fields obtained from FEM and ANSYS showed consistent physical behavior, with maximum values occurring near the fixed support. Overall, the finite element model successfully captures both bending and shear effects, making it a reliable tool for analyzing beam behavior under static and dynamic conditions.